MSE 221 Final Project:
Strength Analysis and Design of a Bicycle Frame

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Abstract

The bicycle frame is the core of the design and is what keeps it together. Because of this, it is important to ensure the frame is capable of withstanding the stresses and forces that will be applied to it. Our chosen bike model, the Marin Terra Linda 3 2021, was assessed and analyzed to see how structurally sound the frame is. Then, a simulation was run in SolidWorks to see the stresses on the frame. After this process, another simulation was run but with increased diameter and thickness in the beams. This change was determined through the stress formula. This proved to be effective in reducing the stresses in the frame and thus increasing the overall strength of the frame.

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Introduction

Transportation has become an incredibly vital part of everyday life. Since the beginning of the Industrial Revolution, and even before then, humanity has been developing more and more innovative products surrounding transportation, eventually leading to the creation of many automotive and manual vehicles. The car, for example, was and still is a fantastic and effective way to get to a desired location quickly. However, that required some sort of fuel, and sourcing the funds to continuously meet the quota to supply the vehicle would become an issue for some. Introduce the bicycle, first invented in the 1800s then developed and improved into the sort of manually operated two-wheeled machine people know today. It has proved itself to be an incredibly energy-effective and long-lasting piece of equipment, being used worldwide by nearly everyone. Keeping the bicycle in that state of longevity would become a challenge that many designers and engineers must tackle, however. In modern designs of any apparatus, research and testing must be done to see what materials and designs work the best for that certain machine, the bicycle being no exception. Choosing a material for the frame of the bicycle, or in other words, the body of the bicycle would be essential in upholding the bicycle's endurance over time. This report will dive deeper into the capabilities of a particular aluminum material suitable for the bike frame of the Marin Terra Linda 3 2021, testing its strength against stresses and forces and comparing those values against improved variations of said bicycle frames. Furthermore, it will discuss the safety implications of the values taken from stress tests, and analyze the individual trusses to propose changes to improve the bicycle's safety and effectiveness. In the *Frame Fatigue Strength* section, the endurance limit of the material was calculated and discussed, followed by calculations and tables done in the *Truss Structure Calculations* section. The frame has been then tested in Solidworks and the tables are discussed in further detail in the following *Discussion of Original Frames Results* section and *Frame Analysis and Faults* section, and then improvement suggestions are noted in the *Proposed Modifications* section. Those improvements were also tested and tabulated, then differences were further discussed in the *Comparison of Original and Modified Bike Frames* section. Finally, the conclusion covers the concepts and reflections done on the tests.

2. Frame Fatigue Strength

S_{FL} = fatigue stress

S_f' = intercept ⇒ 82 ksi

b = slope ⇒ −0.102

NFL = # of loading cycles ⇒ 3000

Aluminum 6061-T6

$$S_{FL} = S_f'(NFL)^b$$

$$S_{FL} = (82)(3000)^{-0.102}$$

$$S_{FL} = 36.2366\,ksi$$

$$\boxed{S_{FL} = 249.8425\,MPa}$$

Default

- S_u = 310.275 MPa
- E = 72397.5 MPa
- $\boxed{S_{FL} = 250\text{ MPa}}$
- S_f' = 565 MPa
- k_{SF} = 0.970
- k_L = 0.923
- k_{size} = 1.000

After calculating the fatigue stress (endurance limit) of aluminum 6061-T6, it is not surprising that an overbearing amount of stress must be applied to cause the material to hit failure under low-cycle fatigue. This calculation, done at 3000 cycles, was also verified on the eFatigue website linked inside the project description to a rounded value of 250 MPa. With these values in mind, the material should be sturdy enough to be used under heavy loads in small cycles in the context of riding a bicycle, as reaching such an amount of pressure whilst using the bicycle would be unrealistic.

4. a) Truss Structure Calculations

Here we can assume each joint is a pin joint with the entire structure being a truss.
The calculations show the Stress, Force and Safety Factors of the mentioned system.

Beam	Stress (MPa)	Force (N)	Safety Factor
AB	11.178	1501.2 (Compression)	23.44
AE	7.453	1001 (Tension)	35.154
BC	6.412	861.18 (Compression)	40.86
BE	3.659	491.35 (Compression)	71.604
EC	5.962	800.68 (Tension)	43.945
ED	1.582	212.49 (Tension)	165.613
CD	5.245	704.46 (Compression)	49.952

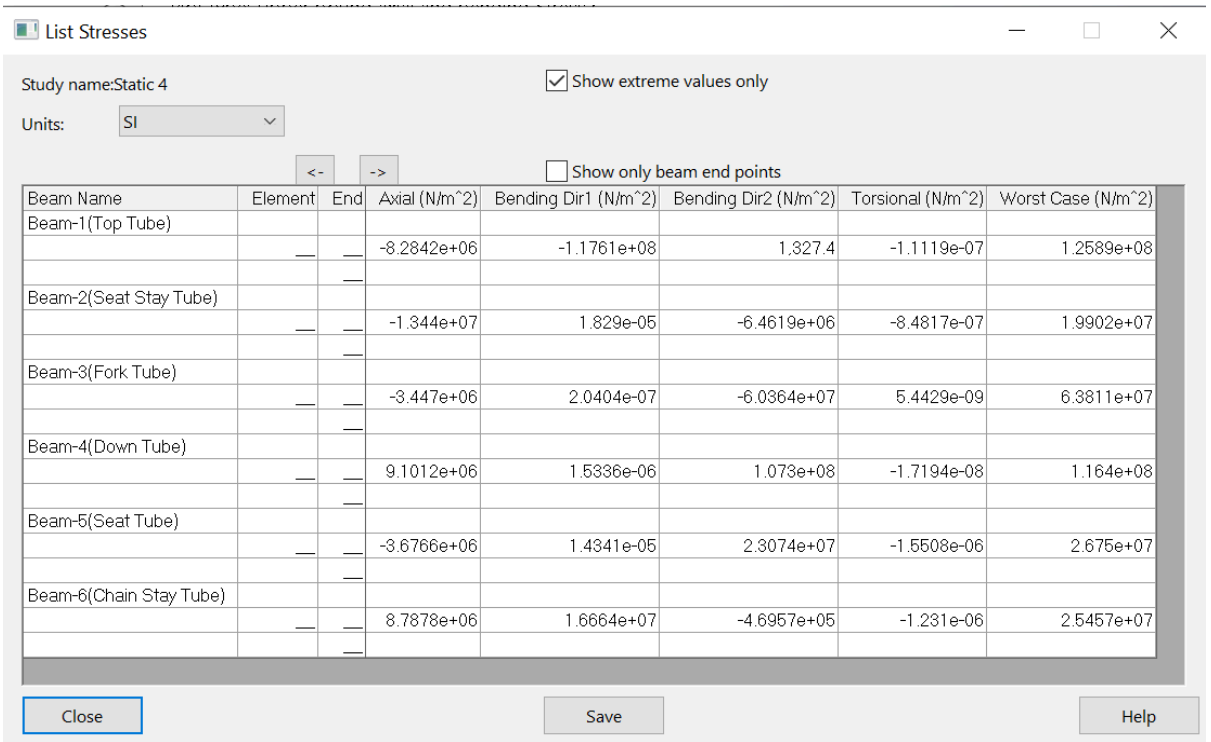
4. b) Combined Truss-Beam Structure Calculations

In this calculation, we assume the top tube of the bike is removed, and that the seat stay and the chainstay are the only two trusses of the structure with the rest being beams.
The calculations show the Stress, Force and Safety Factors of the mentioned system.

Beam	Stress (MPa)	Force (N)	Safety Factor
AB	13.146	1765.51 (Compression)	19.93
AE	8.766	1177.24 (Tension)	29.888
BE	0.86	115.54 (Compression)	304.65
EC	1.4	188.2 (Compression)	187
CD	3.756	504.43 (Compression)	69.8

4. c) SolidWorks FEA Analysis of Original Frame

In this case, we assume all the bars are beams and use a Finite Element Analysis (FEA) to create this table



Beam Name	Element	End	Axial (N/m^2)	Bending Dir1 (N/m^2)	Bending Dir2 (N/m^2)	Torsional (N/m^2)	Worst Case (N/m^2)
Beam-1(Top Tube)							
	__	__	-8.2842e+06	-1.1761e+08	1.327.4	-1.1119e-07	1.2589e+08
		__					
Beam-2(Seat Stay Tube)							
	__	__	-1.344e+07	1.829e-05	-6.4619e+06	-8.4817e-07	1.9902e+07
		__					
Beam-3(Fork Tube)							
	__	__	-3.447e+06	2.0404e-07	-6.0364e+07	5.4429e-09	6.3811e+07
		__					
Beam-4(Down Tube)							
	__	__	9.1012e+06	1.5336e-06	1.073e+08	-1.7194e-08	1.164e+08
		__					
Beam-5(Seat Tube)							
	__	__	-3.6766e+06	1.4341e-05	2.3074e+07	-1.5508e-06	2.675e+07
		__					
Beam-6(Chain Stay Tube)							
	__	__	8.7878e+06	1.6664e+07	-4.6957e+05	-1.231e-06	2.5457e+07
		__					

Fig. 1 - Beam Stress Data, Max Stress = -1.344E7

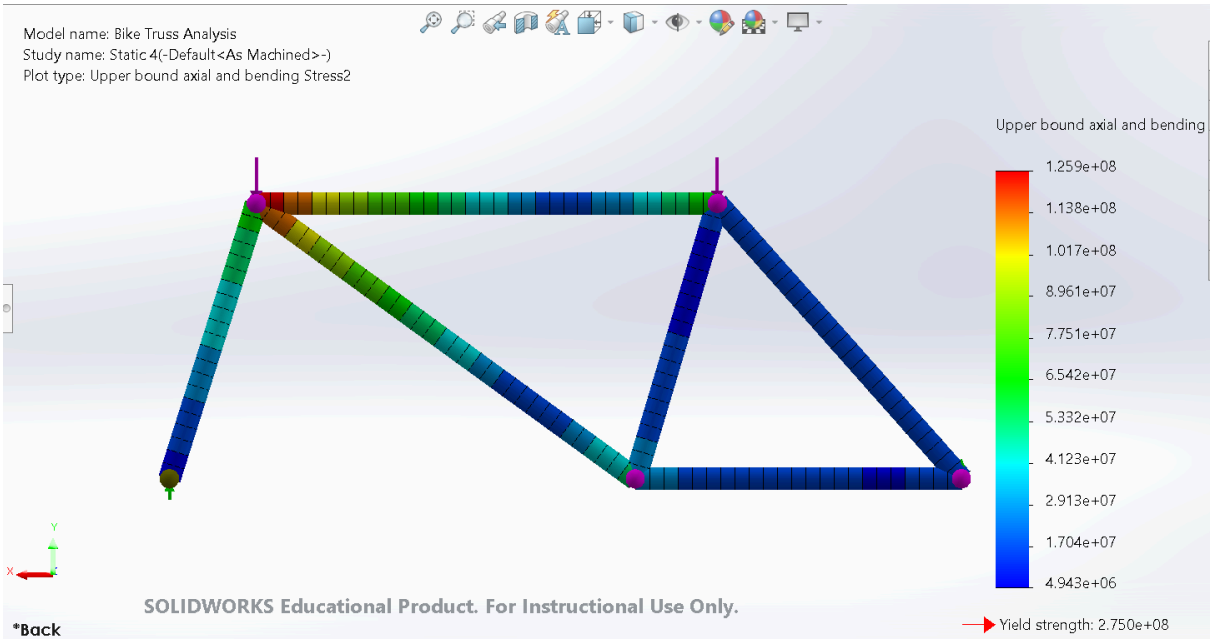


Fig. 2 - Beam Axial and Bending Stress Diagram, Max. Axial/Bending = 1.26E8

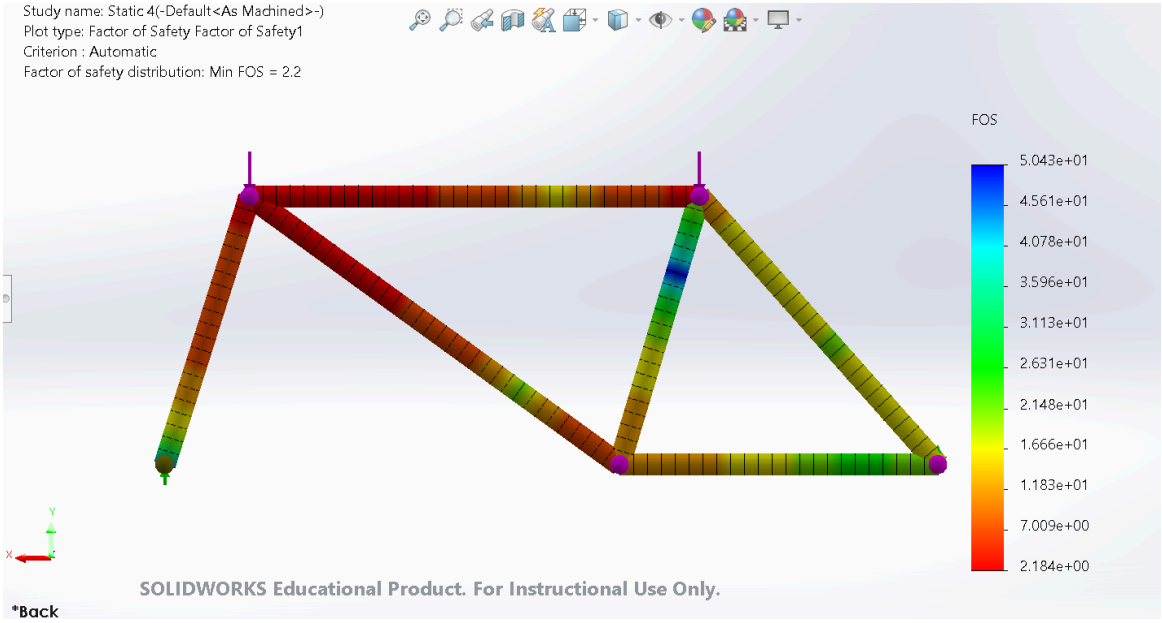


Fig. 3 - Factor of Safety Diagram, Min. FS = 2.184

5. Discussion of Frame's Results

Table 4a lists the stresses, forces, and safety factors in each bar while accounting for the additional bar (that runs from the crank center to the center of the front wheel, ED) that allows us to make this a legitimate truss structure. Table 4b also calculates the stresses, forces, and safety factors of the system but in this case, there is no extra bar (ED) or the top tube (BC). Another thing we account for in this case is that the seat stay(AB) and chain stay(AE) are considered bars while the rest are considered beams. Comparing the tables we can see that the stresses in Table 4b are greater than 4a for the bars AB and AE, this is because all the stress in our system is directed to those two beams and results in the factor of safety being lower in Table 4b. On the other hand beams BE, EC, CD have less stress and therefore result in higher safety factors than 4a. Table 4c assumes all bars are beams which is the real case, using solidworks (FEA) we can simulate the stresses. We can see that the seat stay and chain stay stresses in Table 4c closely resemble the stresses in Table 4b while the other stresses are not similar, this error could either be related to calculation or simulation.

6. Frame Analysis and Faults

a. This frame is safe under the given loading conditions (Fa~ 1600N, Fb~ 200N) (safety factor > 3)
b. Our most risky bar is the seat stay bar (AB) with a stress of -1.344E7
c. According to the stress formula (F/A), we can improve the design by increasing the diameter and thickness of the beams, in turn making them stronger and reducing stress
d. If the weight is reduced we can again refer to the stress formula and reduce the area of the beams, this will save material and make our bike lighter but since it is a cruiser it won't take much effect. Having a lighter bike makes it worth taking a look at how portable we can make it for example making it foldable could help people who commute by train and bike fit their bikes in trains, it can also help people going on small camping trips fit their bikes in the car and leave roof space for other camping necessities.

7. Proposed Modifications

Our proposed design change is based on the stress formula. We can increase the strength of the bars by increasing their dimensions. The bar will have the diameter changed from 30 to 50 millimeters and the thickness changed from 1.5 to 2 millimeters. The change in the bar's dimensions will increase the bike's weight, but since this is a cruiser bike, the handling will not be affected by the increase in weight. Another advantage of this modification is that it preserves the bike's original geometry so the bike will fit the person more naturally and handle similarly.

8. SolidWorks FEA Analysis of Modified Frame

List Stresses

Study name:Static 4

Units: SI

☑ Show extreme values only

☐ Show only beam end points

Beam Name	Element	End	Axial (N/m^2)	Bending Dir1 (N/m^2)	Bending Dir2 (N/m^2)	Torsional (N/m^2)	Worst Case (N/m^2)
Beam-1(Top Tube)							
	__	__	-3.6166e+06	-3.0708e+07	347.03	-2.1241e-08	3.4325e+07
		__					
Beam-2(Seat Stay Tube)							
	__	__	-5.9209e+06	-6.8002e-06	-2.6854e+06	-1.1983e-06	8.6063e+06
		__					
Beam-3(Fork Tube)							
	__	__	-1.5351e+06	-1.0479e-07	-1.581e+07	-9.854e-09	1.7345e+07
		__					
Beam-4(Down Tube)							
	__	__	3.9784e+06	2.2555e-06	2.8201e+07	8.1682e-08	3.2179e+07
		__					
Beam-5(Seat Tube)							
	__	__	-1.6491e+06	6.1871e-06	5.9737e+06	-1.2002e-06	7.6228e+06
		__					
Beam-6(Chain Stay Tube)							
	__	__	3.8553e+06	3.197e+06	-91.371	-1.4624e-06	7.0529e+06
		__					

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Fig. 4 - Upgraded Beam Stress Data, Max. Stress = -5.92E6

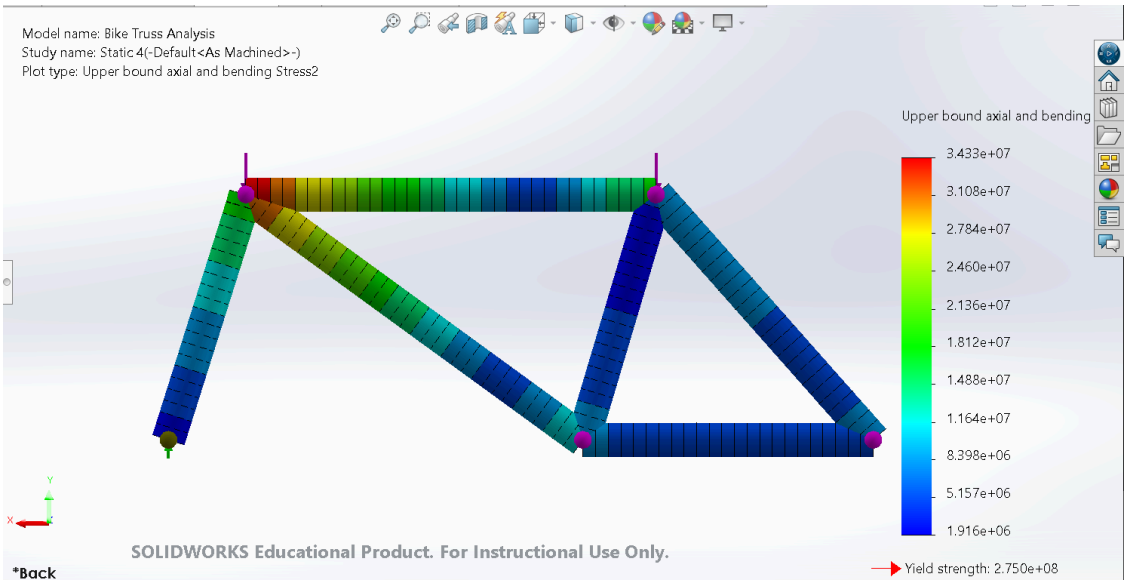


Fig. 5 - Upgraded Beam Stress Diagram, Max. Axial/Bending = 3.43E7

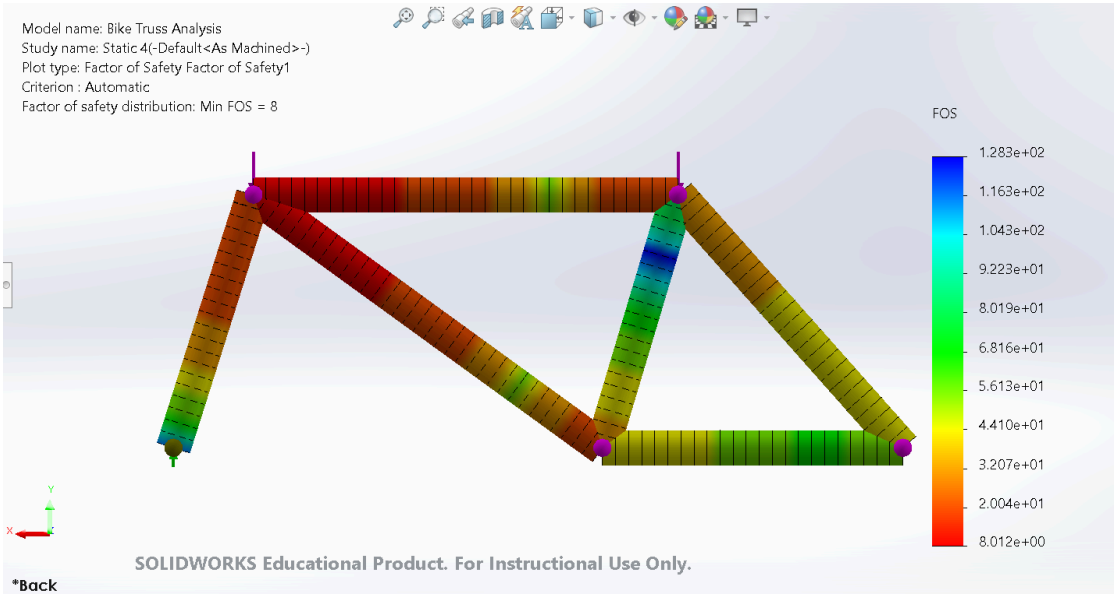


Fig. 6 - Upgraded Factor of Safety Diagram, Min. FS = 8.01

9. Comparison of Original and Modified Bike Frames

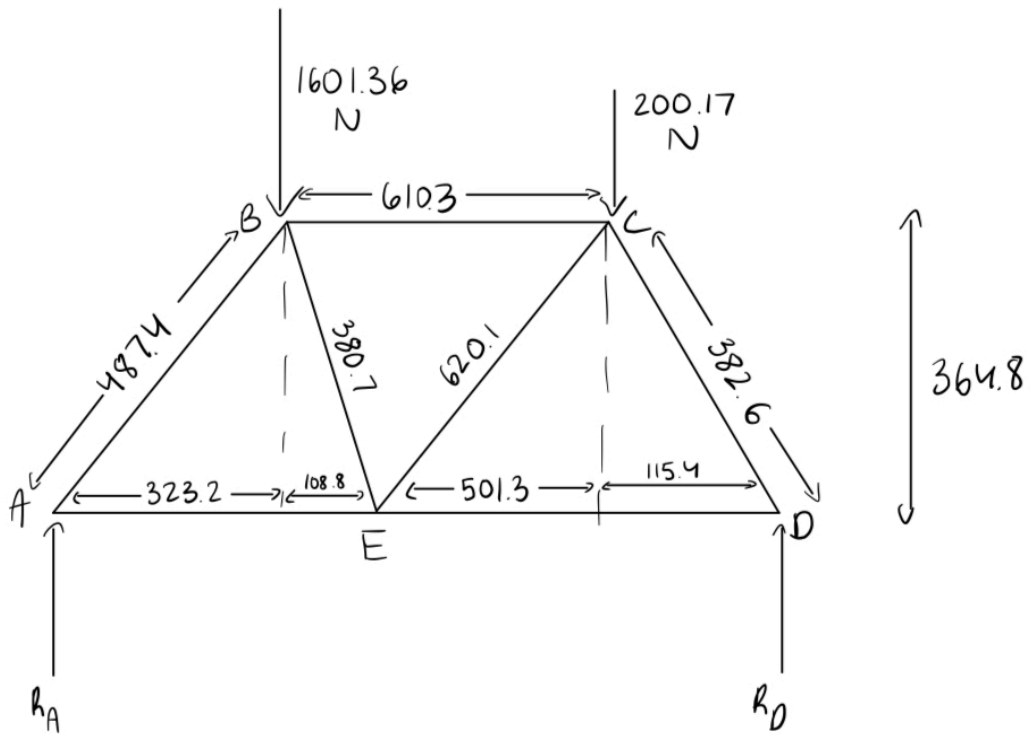
After looking at the simulations we can conclude that the new design is better. As mentioned before, the added weight is negligible, while adding more strength and stiffness to the bike. Furthermore, the simulation also shows a higher safety factor in the new design. This is illustrated in the increase in the lowest safety factor going from 2.2 to 8.0. However, there is a possibility of error in the process of creating the simulation in SolidWorks such as the design itself or wrong parameters. Regardless, the stress formula is force over area which would mean having more area will reduce the stress applied to the beams. This is consistent with our results and is the basis of the idea. If we were to design it again, we would follow the same idea of increasing the beam's dimensions. The idea worked well and could be further worked upon to achieve more desirable results, although alternative improvements might be possible.

Conclusion

This project furthered our knowledge of how stresses and forces can affect a truss-like system, especially how they can be split between bars and how angles affect them. We also now have a greater understanding of the important roles, materials, and design play in creating a safe truss structure. We started by making measurements and calculations of our sample bike, creating simulated scenarios with the different forces stresses and components of the structure as proof of concept of our calculations. The results from these simulations helped us deduce questions we had later on in our project, such as the safety factor and riskiest part of the bike. The results also helped set us up for further innovation, such as suggestions for improvement, changes we would make under different conditions, and proposing a design change and then analyzing it to see if it was for the better or worse. In our case, our design change included increasing the dimensions of the bar (diameter, thickness, and area). We found this to be the best change as it adds strength to the structure while preserving the user-friendly design of the bike. Through a comparison of our pre- and post-design change calculations, we were able to deem that our new design was better as we were able to add strength and sturdiness while adding a negligible amount of weight and increasing our safety factor.

Appendix

Free body diagram of the entire truss structure for the calculations in 4a. Includes the points which will be referenced more below.



Calculations for the reaction forces and isolating joint A:

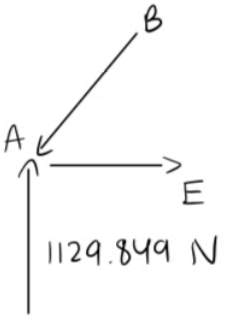
$$\Sigma M_A = 0 \quad -(1601.36)(0.3232) - (200.17)(0.9333) + R_D(1.0487) = 0$$

$$R_D = 671.681 \text{ N}$$

$$\Sigma F_y = 0 \qquad -1601.36 - 200.17 + 671.681 + R_A = 0$$

$$R_A = 1129.849 \text{ N}$$

Point A



$$\frac{364.8}{484.7}(F_{AB}) = 1129.849 \text{ N}$$

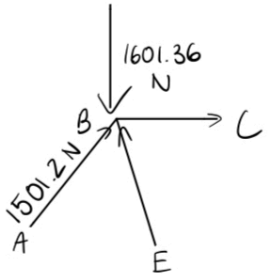
$$F_{AB} = 1501.2 \text{ N} \qquad \text{Compression}$$

$$\left(\frac{323.2}{484.7}\right) F_{AB} = F_{AE}$$

$$1001 \text{ N} = F_{AE} \qquad \text{Tension}$$

Isolating joint B:

Point B



$$\left(\frac{364.8}{487.4}\right) F_{AB} + \left(\frac{364.8}{380.7}\right) F_{BE} = 1601.36$$

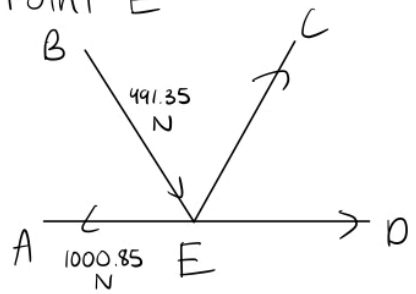
$$F_{BE} = 491.35 \text{ N} \qquad \text{Compression}$$

$$\left(\frac{323.2}{487.4}\right) F_{AB} - \left(\frac{108.8}{380.7}\right) F_{BE} + F_{BC} = 0$$

$$F_{BC} = 861.18 \text{ N} \qquad \text{compression}$$

Isolating joints E and D:

Point E

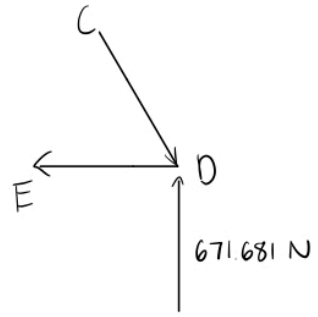


$$\left(\frac{364.8}{620.1}\right)F_{EC} - \left(\frac{364.8}{380.7}\right)F_{BE} = 0$$

$$\underline{F_{EC} = 800.68\ N}$$ Tension

$$\sum F_x = 0 \quad -1000.85 + \left(\frac{108.8}{380.7}\right)F_{BE} + \left(\frac{501.3}{620.1}\right)F_{EC} + F_{ED} = 0$$

Point D



$$\left(\frac{115.4}{382.6}\right)F_{CD} = \underline{F_{ED} = 212.49\ N}$$ Tension

$$\frac{364.8}{382.6}(F_{CD}) = 151$$

$$\underline{F_{CD} = 704.46\ N}$$ compression

The calculations for finding the area of a singular beam as well as the stresses in every individual beam The value for the ultimate stress was found in the textbook.

$$A = \frac{\pi(0.03)^2}{4} - \frac{\pi(0.027)^2}{4}$$

$$= 1.343 \times 10^{-4}\ m^2$$

$\sigma_{Al} = 38\ ksi = 262\ MPa$ from textbook

Beam AB: $\sigma_{AB} = \frac{1501.2\ N}{1.343 \times 10^{-4}\ m} = 11.178\ MPa$

$$SF_{AB} = \frac{262}{11.178} = 23.44$$

Beam AE: $\sigma_{AE} = \frac{1001\ N}{1.343 \times 10^{-4}\ m} = 7.453\ MPa$

$$SF_{AE} = \frac{262}{7.453} = 35.15$$

Beam BC: $\sigma_{BC} = \frac{861.18\ N}{1.343 \times 10^{-4}\ m} = 6.412\ MPa$

$$SF_{BC} = \frac{262}{6.412} = 40.86$$

Beam BE: $\sigma_{BE} = \frac{491.35\ N}{1.343 \times 10^{-4}\ m} = 3.659\ MPa$

$$SF_{BE} = \frac{262}{3.659} = 71.604$$

Beam EC: $\sigma_{EC} = \frac{800.68\ N}{1.343 \times 10^{-4}\ m} = 5.962\ MPa$

$$SF_{EC} = \frac{262}{5.962} = 43.94$$

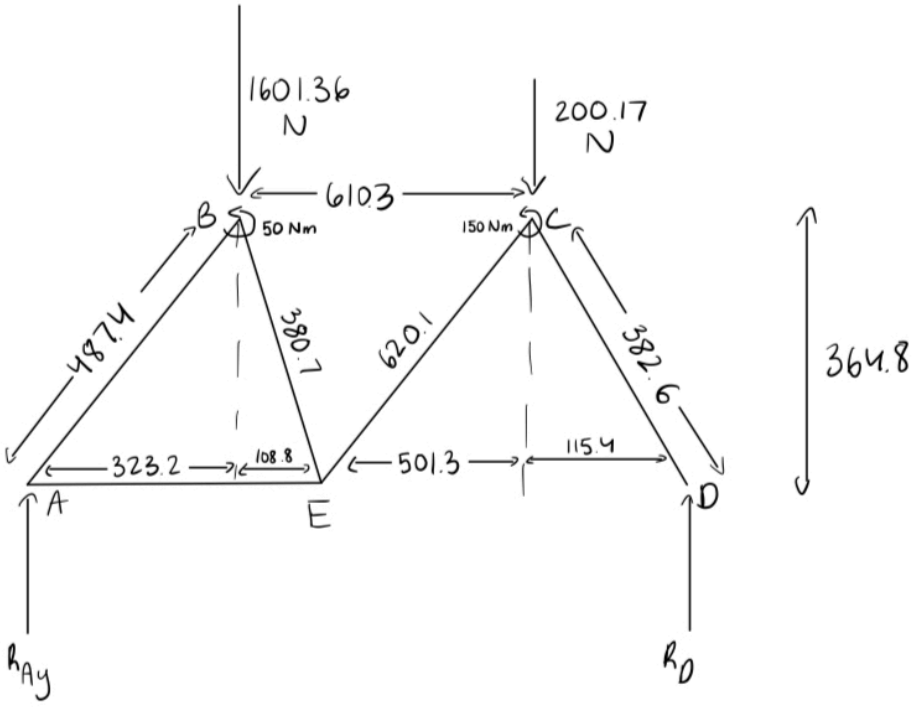
Beam ED: $\sigma_{ED} = \frac{212.49\ N}{1.343 \times 10^{-4}\ m} = 1.582\ MPa$

$$\frac{262}{1.582} = 165.61$$

Beam CD: $\sigma_{CD} = \frac{704.46\ N}{1.343 \times 10^{-4}\ m} = 5.245\ MPa$

$$\frac{262}{5.245} = 49.95$$

Below shows the free body diagram for beam and truss structure for part 4b. Two additional moments were added, one on joint B and one on joint C. Their respective values are 50 Nm and 150 Nm, which were used for calculating the reaction forces.

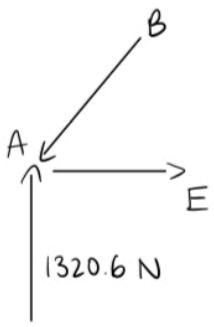


$$\sum M_A = 0 \quad -(1601.36)(0.3232) - (200.17)(0.9333) + R_D(1.0487) + 50 + 150 = 0$$

$$\underline{R_D = 480.96 \text{ N}}$$

$$\sum F_y = 0 \quad -1601.36 - 200.17 + 480.96 + R_{AY} = 0 \qquad R_{AY} = 1320.6 \text{ N}$$

Point A



$$\frac{364.8}{484.7}(F_{AB}) = 1320.6$$

$$\underline{F_{AB} = 1765.51 \text{ N}} \qquad \text{Compression}$$

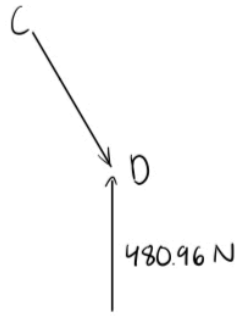
$$\left(\frac{323.2}{484.7}\right) F_{AB} = F_{AE}$$

$$\underline{1177.24 \text{ N} = F_{AE}} \qquad \text{Tension}$$

Above shows isolating point A.

Below shows the calculations for isolating point C, D, and E

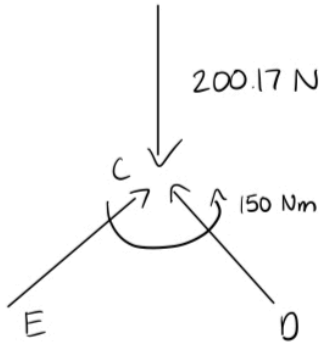
Point D



$$\frac{364.8}{382.6}(F_{CD}) = 480.96\ N \qquad \underline{F_{CD} = 504.43\ N}$$

Compression

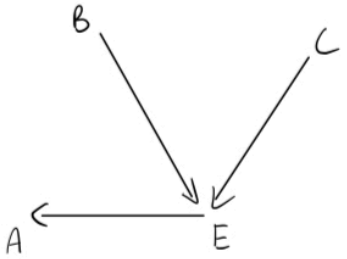
Point C



$$\left(\frac{501.3}{620.1}\right)F_{EC} - \left(\frac{115.4}{382.6}\right)F_{CD} = 0 \qquad \underline{F_{EC} = 188.2\ N}$$

Compression

Point E



$$\frac{364.8}{620.1}F_{EC} = \frac{364.8}{380.7}F_{BE}$$

$$\underline{F_{BE} = 115.54\ N}$$

Compression

Once again, the calculations below show the process for finding the stresses in each beam for the structure in 4b.

$$A = \frac{\pi(0.03)^2}{4} - \frac{\pi(0.027)^2}{4}$$

$$= 1.343 \times 10^{-4}\ m^2$$

$\sigma_{Al} = 38$ ksi $= 262$ MPa
from textbook

Beam AB: $\sigma_{AB} = \frac{1765.51\ N}{1.343 \times 10^{-4}\ m} = 13.146$ MPa

$$SF_{AB} = \frac{262}{13.146} = 19.93$$

Beam AE: $\sigma_{AE} = \frac{1177.24\ N}{1.343 \times 10^{-4}\ m} = 8.766$ MPa

$$SF_{AE} = \frac{262}{8.766} = 29.888$$

Beam BE: $\sigma_{BE} = \frac{115.54\ N}{1.343 \times 10^{-4}\ m} = 0.86$ MPa

$$SF_{BE} = \frac{262}{0.86} = 304.65$$

Beam EC: $\sigma_{EC} = \frac{188.2}{1.343 \times 10^{-4}\ m} = 1.4$ MPa

$$SF_{EC} = \frac{262}{1.4} = 187$$

Beam CD: $\sigma_{CD} = \frac{504.43}{1.343 \times 10^{-4}\ m} = 3.756$ MPa

$$\frac{262}{3.756} = 69.8$$